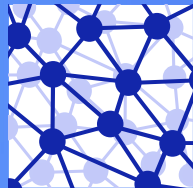


Explainable AI

DeNederlandscheBank

EUROSYSTEEM



DataScience
Hub

Workshop content

- Why explainable AI?
- Practical Explainable AI:
 - Data exploration
 - White box models
 - Global feature importance
 - Local importance with SHAP

Introduction

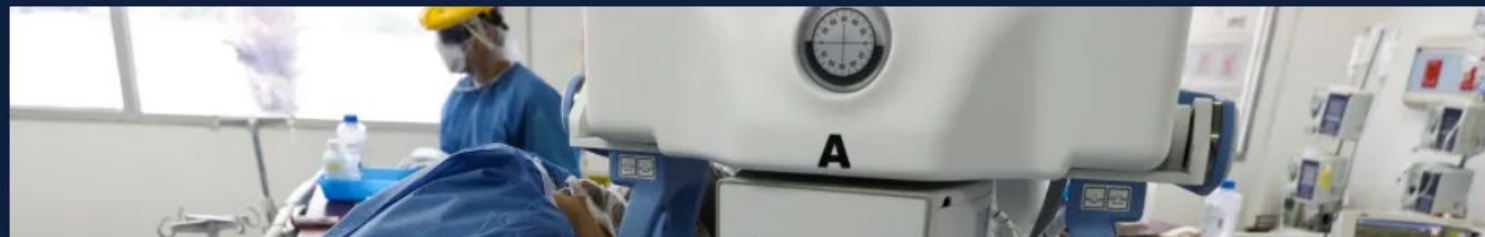
ARTIFICIAL INTELLIGENCE

Hundreds of AI tools have been built to catch covid. None of them helped.

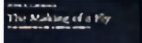
Some have been used in hospitals, despite not being properly tested. But the pandemic could help make medical AI better.

By Will Douglas Heaven

July 30, 2021



Introduction



The Making of a Fly: The Genetics of Animal Design (Paperback)

by Peter A. Lawrence

[Return to product information](#)

Always pay through Amazon.com's Shopping Cart or 1-Click.
Learn more about [Safe Online Shopping](#) and our [safe buying guarantee](#).

Price at a Glance

List Price: ~~\$70.00~~
Used: from **\$42.56**
New: from **\$18,651,718.08**
Have one to sell? [Sell yours here](#)

Years 1922 to 1936 (4) **\$14²³** 

Condition 

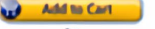
Buy **Deliv**
Or fa
8. On
 De
Used
Sold |
Fulfil
Access
guarai

Adc

All **New** (2 from \$18,651,718.08) **Used** (11 from \$42.56)

Show   **Prime offers only (0)** Sorted by **Price + Shipping**

New 1-2 of 2 offers

Price + Shipping	Condition	Seller Information	Buying Options
\$18,651,718.08 + \$3.99 shipping	New	Seller: profnath Seller Rating:  93% positive over the past 12 months. (8,278 total ratings) In Stock. Ships from NJ, United States. Domestic shipping rates and return policy . Brand new, Perfect condition, Satisfaction Guaranteed.	 or Sign in to turn on 1-Click ordering.
\$23,698,655.93 + \$3.99 shipping	New	Seller: bordeebook Seller Rating:  93% positive over the past 12 months. (127,332 total ratings) In Stock. Ships from United States. Domestic shipping rates and return policy . New item in excellent condition. Not used. May be a publisher overstock or have slight shelf wear. Satisfaction guaranteed!	 or Sign in to turn on 1-Click ordering.

(125,891 total ratings) [Sign in](#) to turn on 1-Click ordering.

In Stock. Ships from United States.
[Domestic shipping rates](#) and [return policy](#).
New item in excellent condition. Not used. May be a publisher overstock or have slight shelf wear. Satisfaction guaranteed!

992  English 

challenges in
nism's genetic
our present
s of this text
primitive
organs and body
juage

Why explainable AI?

- Robustness
- Causality
- Fairness
- Trust



Why explainable AI?

- Robustness
- Causality
- Fairness
- Trust

Wolf vs Husky



Why explainable AI?

- Robustness
- Causality
- Fairness
- Trust

Wolf vs Husky



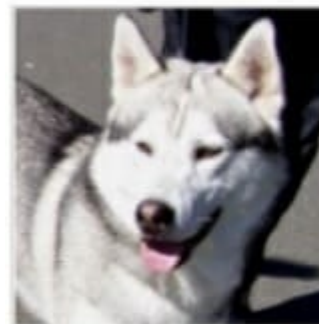
Predicted: **wolf**
True: **wolf**



Predicted: **husky**
True: **husky**



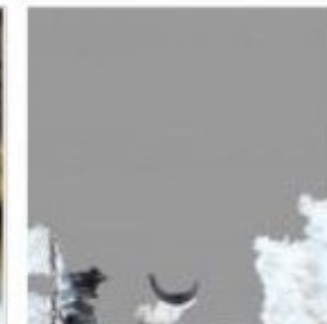
Predicted: **wolf**
True: **wolf**



Predicted: **husky**
True: **husky**



Predicted: **wolf**
True: **wolf**



Predicted: **wolf**
True: **husky**

Why explainable AI?

- Robustness
- Causality
- Fairness
- Trust

RETAIL OCTOBER 11, 2018 / 1:04 AM / UPDATED 4 YEARS AGO

Amazon scraps secret AI recruiting tool that showed bias against women

By Jeffrey Dastin

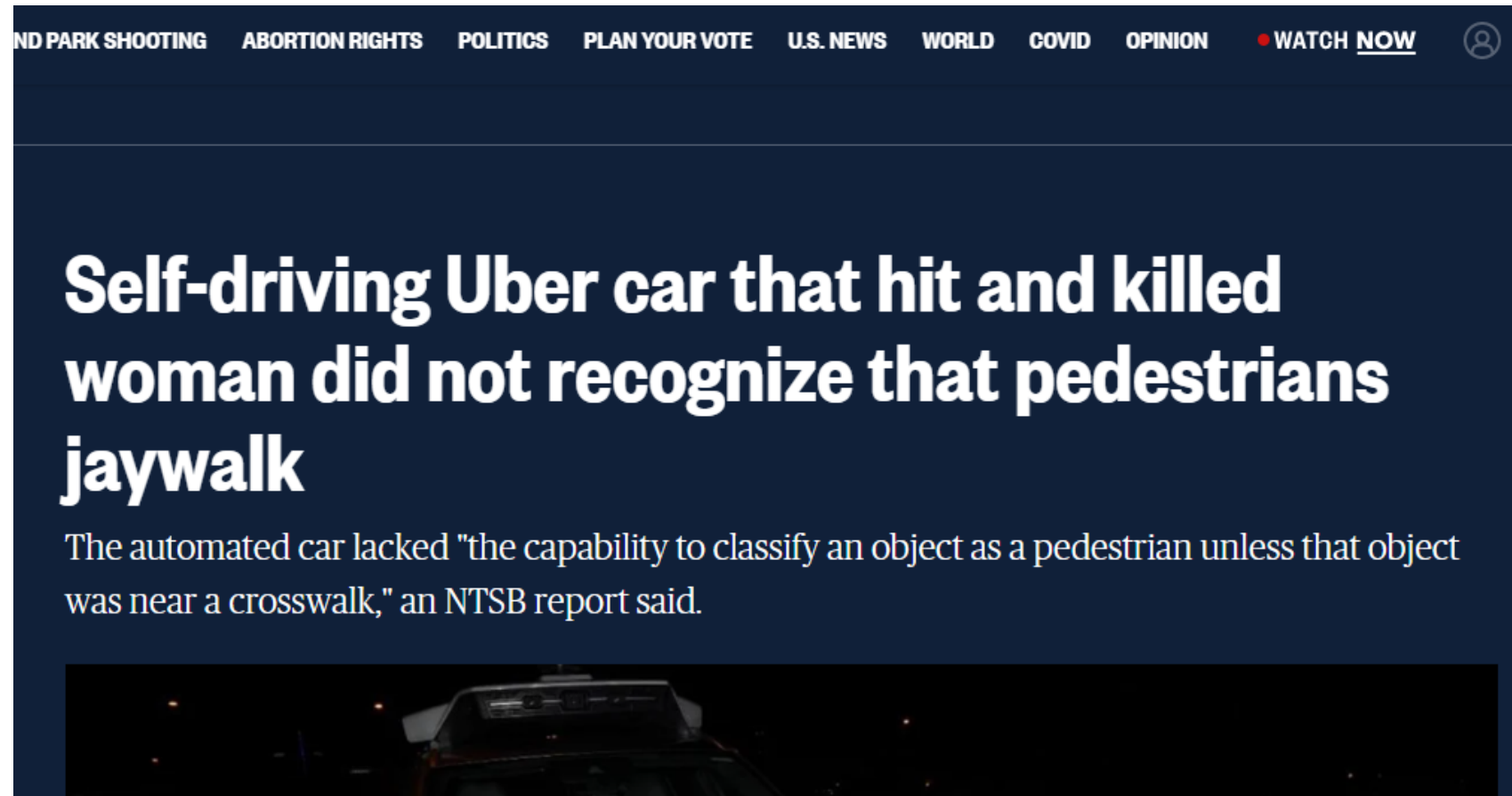
8 MIN READ



SAN FRANCISCO (Reuters) - Amazon.com Inc's [AMZN.O](#) machine-learning specialists uncovered a big problem: their new recruiting engine did not like women.

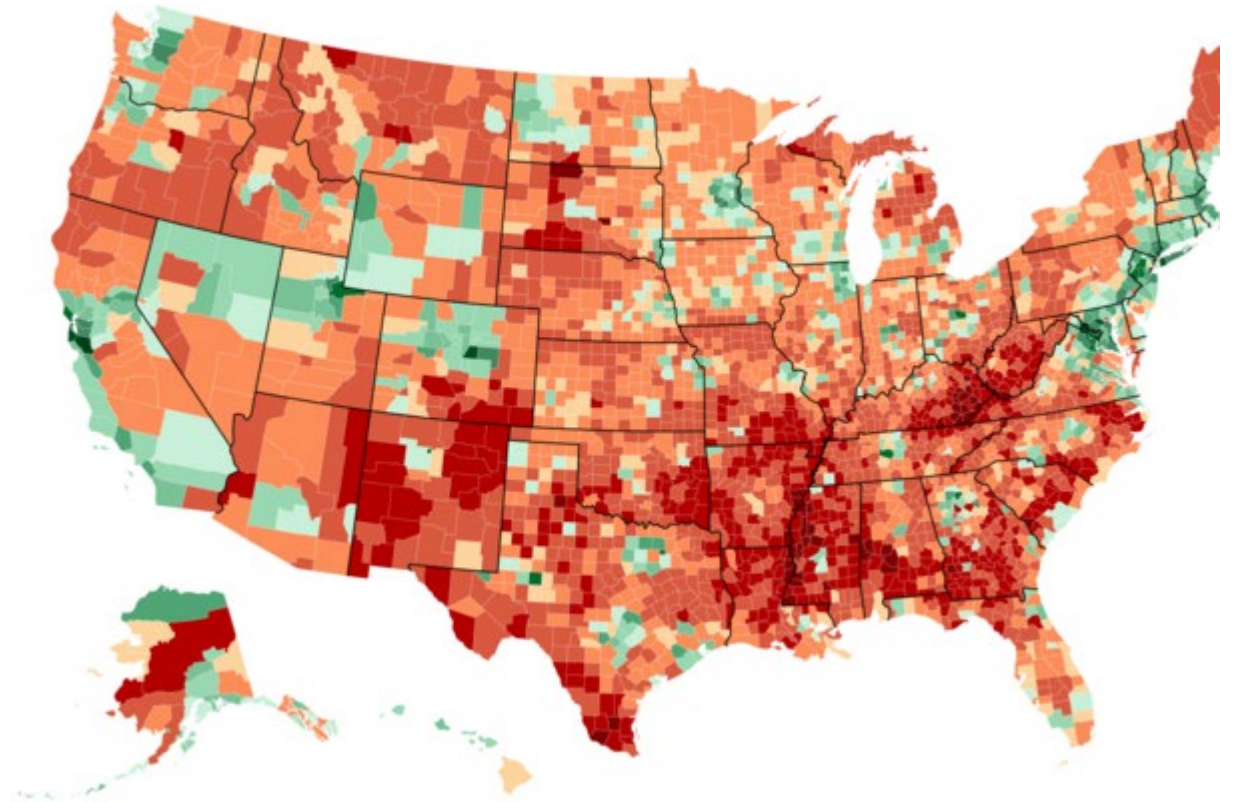
Why explainable AI?

- Robustness
- Causality
- Fairness
- Trust



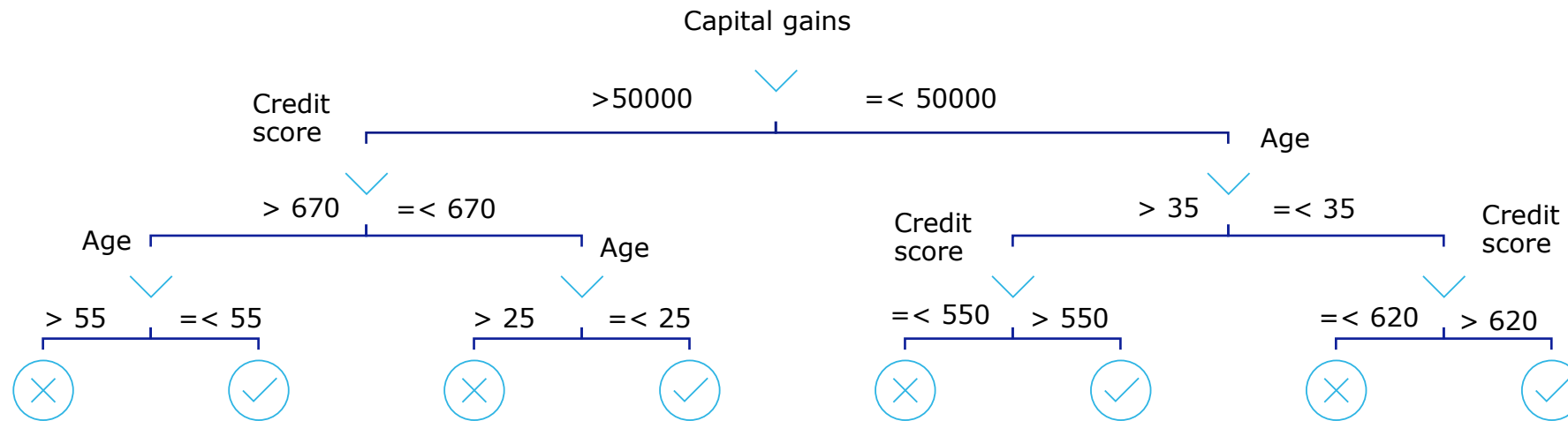
Adult Census-Income Data Set

- Census data extracted from the 1994 and 1995 Current Population Surveys conducted by the U.S. Census Bureau.
- 12 variables such as:
age, sex, capital gains, etc..
- **Aim:**
Predict whether the total person income is larger then \$50.000.



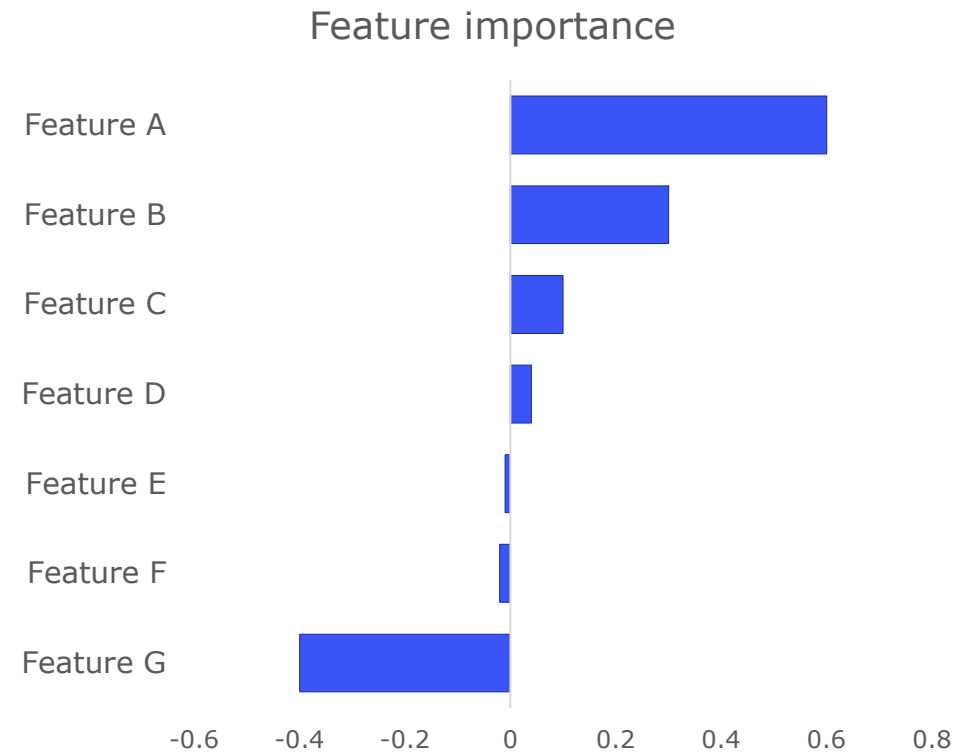
White box models

- **Models which are easy enough to understand**
- Linear regression or decision trees



Global explainability

- **What variables are most important in our model?**
- Global explainability
- *What factors are important for predicting a higher default rate?*
- Permutation & Perturbation -> Partial dependence plots



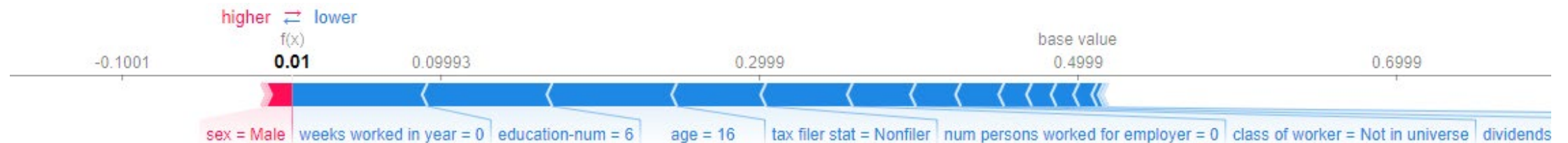
Local explainability

- **Why is a certain decision made?**

Local explainability

Why does loan X have a higher default rate?

LIME, Shapley values (SHAP), Counterfactuals



Shapley values

- Pretend the AI model is a cooperative game
 - Payout is the prediction accuracy
 - Players are the parameters
- Calculate a *fair* way to distribute the payout over the players

$$\phi_i(v) = \frac{1}{|N|} \sum_{S \subseteq N \setminus \{i\}} \left(\frac{n-1}{|S|} \right)^{-1} [v(S \cup \{i\}) - v(S)]$$

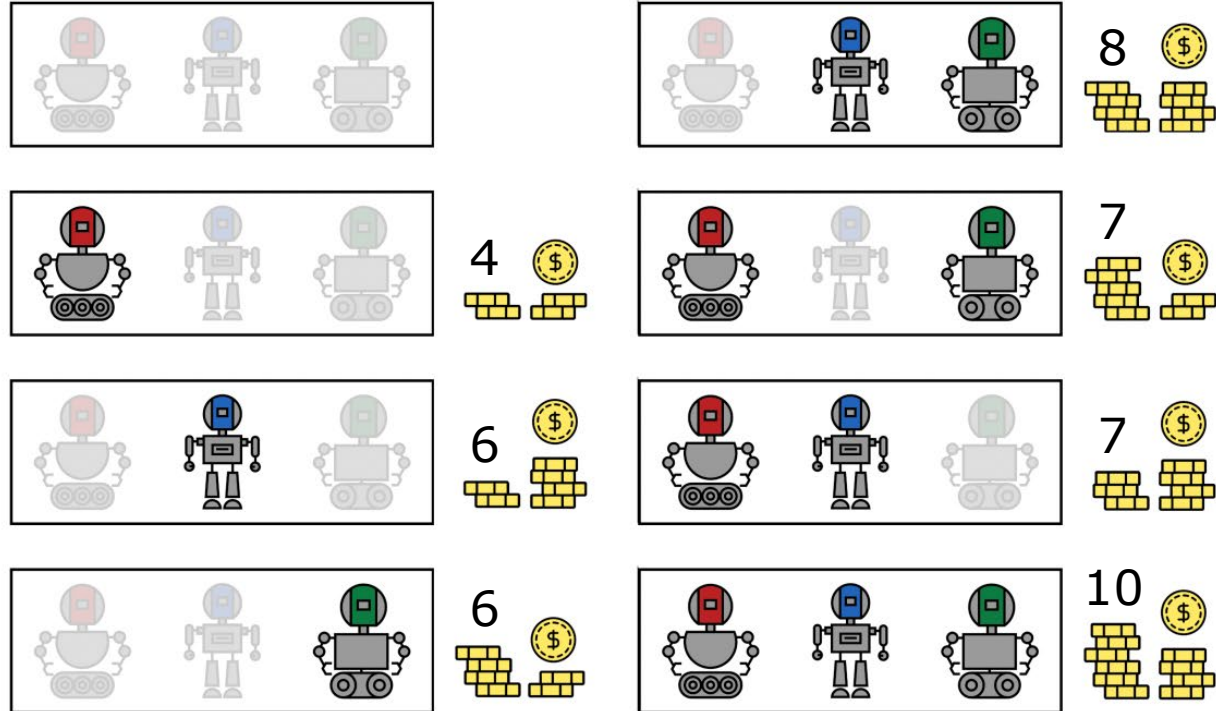
Relative importance of player i

Number of players

Number of coalitions of this size without player i

Additional payout of including i to the coalition

Shapley values



Counterfactuals

